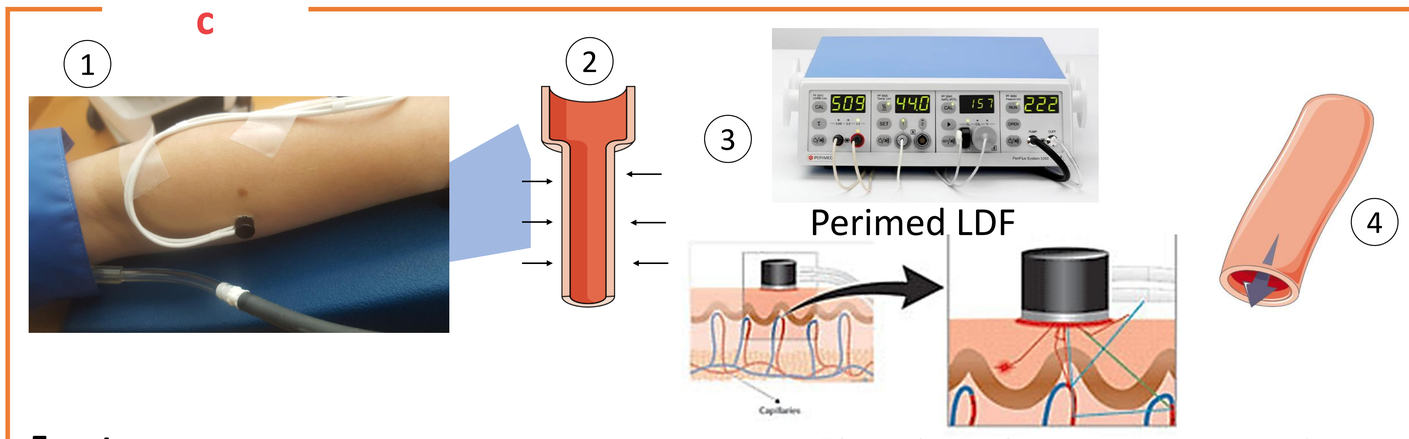
DÉTECTION AUTOMATIQUE D'ÉVÈNEMENTS MICROVASCULAIRES PAR APPRENTISSAGE PROFOND

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INTRODUCTION

Post-occlusive reactive hyperemia (PORH) is a non-invasive method used to assess peripheral microvascular function, which is essential for oxygen and nutrient supply to tissues. The PORH signal obtained from this test is noisy and complex to analyze. The development of new analysis techniques, such as the use of advanced mathematical models or machine learning algorithms, would allow for a more accurate and detailed assessment of PORH signals, thus offering deeper insights into vascular function. These methods are particularly promising for the study of microcirculation. **The objective of this project is to develop a fully automatic pipeline for analyzing PORH signals using hidden Markov models (HMM) and an LSTM neural network for parameter prediction in order to refine the interpretation of PORH recordings.**

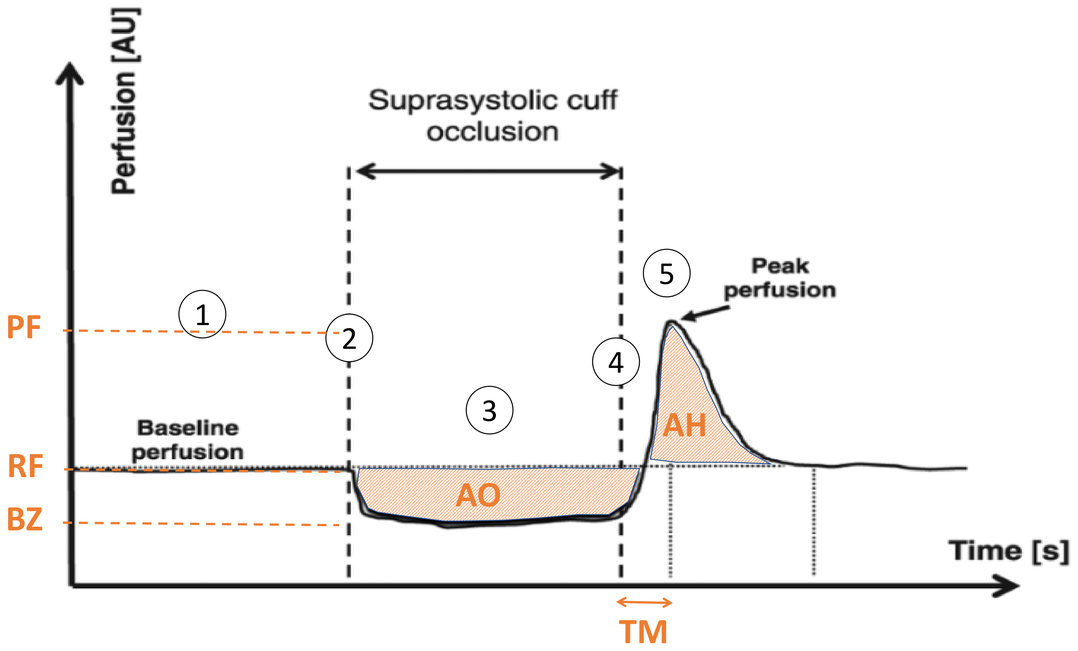


Events :

1- Measurement of Basal Flow at rest.
2- The cuff is inflated above systolic pressure for 5 minutes, causing vascular occlusion of the vessel.
3- Measurement of the Biological Zero.
4- Release the pressure in the cuff to 0 mmHg.
5- Hyperaemia.

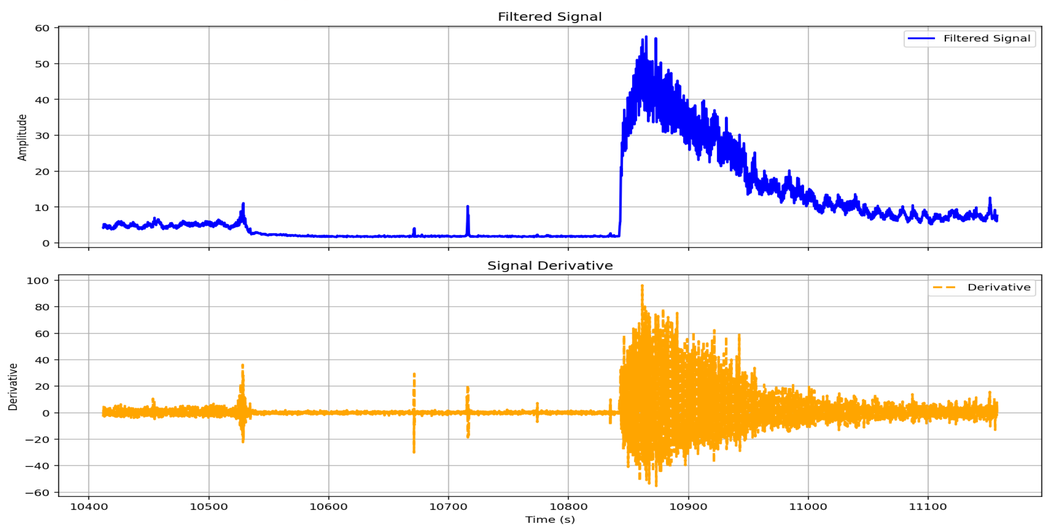
Physiological parameters to predict:

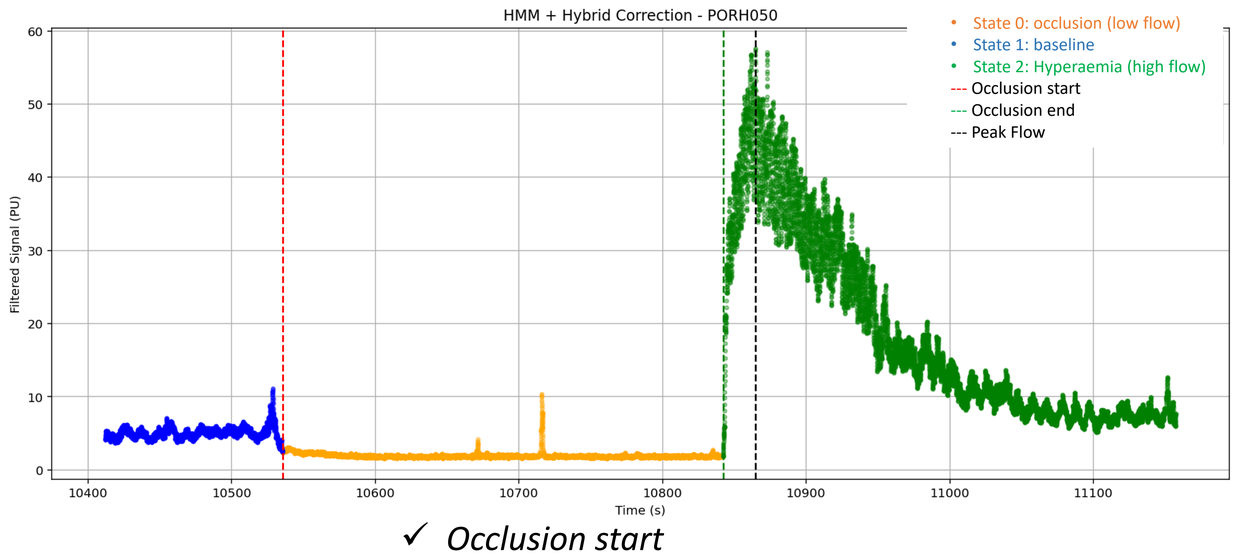
RF: Rest Flow
BZ: Biological Zero
AO: Occlusion Area
PF: Peak Flow
TM: Time to Max (Time-to-Peak)
AH: Hyperaemia Area



96 Recordings from patients with severe obesity

Methodology





A. Preprocessing

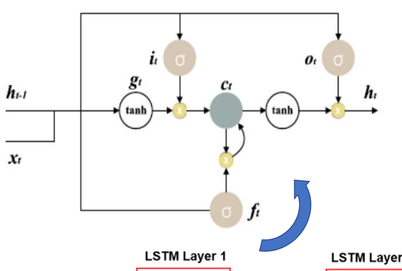
- Time normalization
- Signal Filtering (Rolling median + Savitzky–Golay)
- First Derivative

B. HMM-based phase segmentation

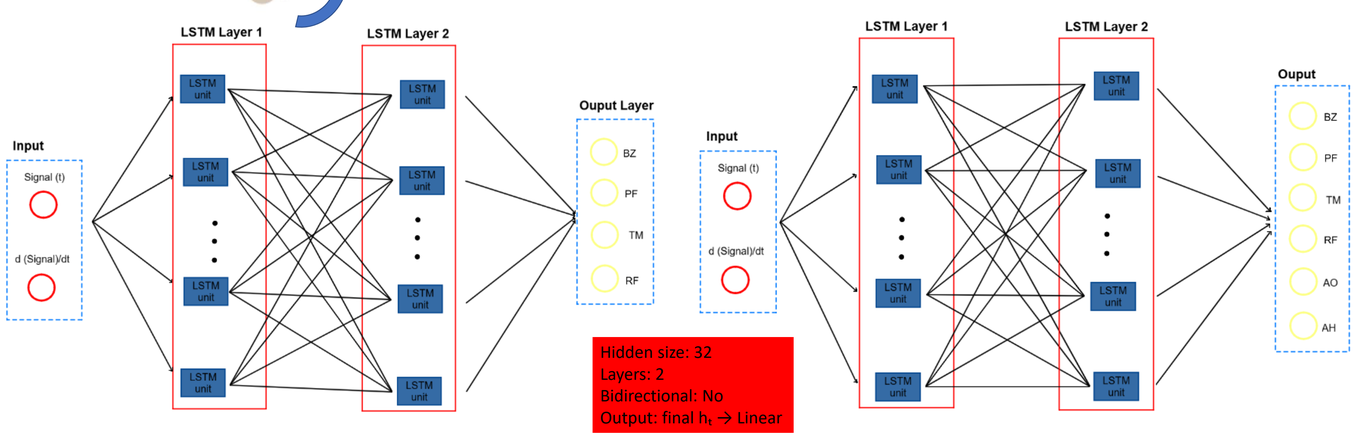
- Features per time point : $X = [S(t), S'(t), |S'(t)|, S''(t)]$
- 3-state gaussian HMM
- Extracted events :
  - ✓ *Occlusion start*
  - ✓ *Occlusion end*
  - ✓ *Peak time*

HMM segmentation provides the event landmarks (baseline, occlusion start, peak) required to compute PORH parameters and to supervise the LSTM regression.

C. LSTM Regression Model



Forget gate f_t: decides how much of the previous memory c_{t-1} is kept.
Input gate i_t: controls how much new information (from x_t, h_{t-1}) enters the cell.
Candidate state c_t: proposed new content added to memory.
Output gate o_t: selects which part of the updated memory is exposed as the new hidden state h_t.
Input at time step t : x_t
Previous hidden state: h_{t-1}



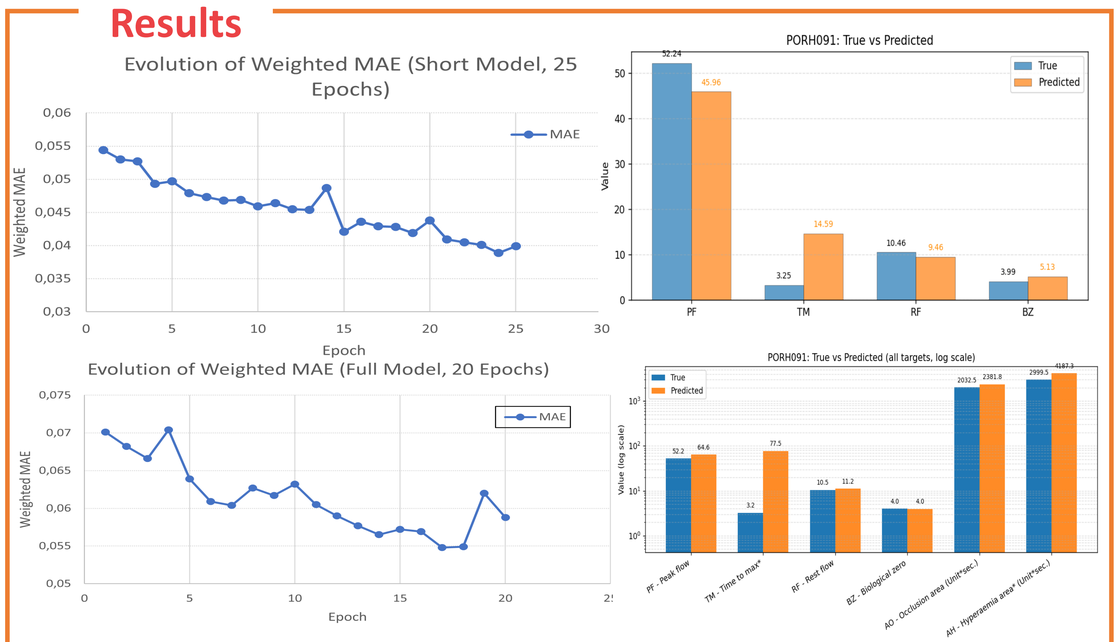
Short Model

Predicts BZ, PF, TM and RF. These parameters depend mainly on **local events** of the signal (baseline, peak and timing). A smaller model is used here to validate that the LSTM can reliably extract key temporal features.

Full Model

Adds AO and AH, which depend on the **entire curve shape**. Requires the LSTM to learn **global temporal dynamics** across baseline–occlusion–hyperaemia. This second model evaluates the network's ability to capture longer-term dependencies.

Results



- The short model shows some convergence and stable learning, reaching a final weighted MAE of ~**0.041** after 25 epochs. Example predictions on PORH091 indicate good accuracy for event-based parameters (PF, TM, RF, BZ), which depend mainly on local temporal features.

- The full model achieves a weighted MAE of ~**0.056** but less converging after 20 epochs. AO and AH, which describe areas under the curve, show larger deviations due to their global dependence on the entire PORH response. Despite this complexity, the model captures all six parameters consistently.

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